

HSE TOOLBOX TALK

GAS TESTING AND ATMOSPHERIC MONITORING

PURPOSE AND SCOPE

Test work atmospheres for oxygen, flammability and toxic contaminants using the correct instrument, sequence and representative locations, maintaining safe decisions as conditions change.

KEY HAZARDS

 1 OXYGEN DEFICIENCY	 2 OXYGEN ENRICHMENT	 3 FLAMMABLE GAS	 4 H2S / CO
 5 WRONG SENSOR	 6 CALIBRATION ERROR	 7 DEAD ZONE	 8 CHANGING ATMOSPHERE

CRITICAL VALUES TO VERIFY IN THE FIELD

19.5-23.5% Oxygen range for permit-space entry	<10% LEL Below hazardous-atmosphere threshold	O2 → LEL → toxic Atmospheric test sequence
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REMEMBER - SERNEM

S

Stop and think

E

Evaluate the risk

R

Remove or reduce

N

Never assume

E

Ensure protection

M

Monitor and communicate

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STANDARDS AND FIELD REQUIREMENTS

OSHA 1910.146 requires defined acceptable entry conditions, atmospheric testing, calibrated instruments and monitoring through entry.

OSHA 1910.146(d)(5)

TESTING AND MONITORING

Test before entry and monitor frequently enough during entry to verify hazardous atmospheres do not develop.

OSHA 1910.146 App B

TEST SEQUENCE

Test oxygen first, then combustible gases/vapours, then potential toxic contaminants.

OSHA 1910.146(b)

HAZARDOUS ATMOSPHERE

Treat oxygen below 19.5% or above 23.5%, and flammables at or above 10% LEL, as hazardous.

ÜRETİCİ / SAHA PROSEDÜRÜ

BUMP, CALIBRATION AND PUMP

Verify daily functional test, calibration status, sensor suitability, hose delay and pump flow.

MINIMUM EVIDENCE BEFORE START

- | | |
|---|--|
| ☐ Gas-detector identification | ☐ Bump / calibration record |
| ☐ Target-gas and limit list | ☐ Pre-entry test results |
| ☐ Continuous-monitoring record | ☐ Ventilation / evacuation plan |

SUPERVISOR DISCUSSION POINTS

- Which gases are we measuring, and why?
- Where could the heaviest gas collect?
- Which route will the team use on alarm?

TOP-MIDDLE-BOTTOM TESTING WITH A CALIBRATED INSTRUMENT



Bump - Calibration - O2 - LEL - Toxic - Continuous monitoring

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CRITICAL CONTROL MEASURES (HIERARCHY OF CONTROLS)

1. ELIMINATE THE HAZARDOUS ATMOSPHERE

Remove product, clean the space and choose a method that does not add contaminants.

2. ISOLATE AND VENTILATE

Isolate lines and mechanically ventilate with clean air; never ventilate with oxygen.

3. TEST REPRESENTATIVE LOCATIONS

Remote-sample top, middle, bottom and pockets, allowing for hose response time.

4. MONITOR CONTINUOUSLY AND EVACUATE

Monitor with personal/fixed instruments; on alarm everyone exits and the area is reassessed.

CRITICAL WARNING



Never silence an alarm and continue work. Stop, exit away from exposure and investigate only from a safe area.

SITE CHECKLIST (BEFORE STARTING WORK)

- | | |
|--|--|
| ☐ Target gases identified | ☐ Sensors suitable |
| ☐ Bump test passed | ☐ Calibration current |
| ☐ Pump / hose sound | ☐ Top-middle-bottom tested |
| ☐ Limits on permit | ☐ Continuous monitoring set |
| ☐ Alarm evacuation understood | |

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ATTENDANCE AND APPROVAL

Project / Site:

Date:

Presented by:

Work area:

No	Full Name	Company / Role	Signature
1			
2			
3			
4			
5			
6			
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12			